Feed-Forward 3D Scene Modeling: A Problem-Driven Perspective

Wang, Weijie, Cao, Qihang, Gao, Sensen, Chen, Donny Y., Xu, Haofei, Bian, Wenjing et al. | cs.CV

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Research Questions

- How can we systematically organize feed-forward 3D reconstruction methods beyond just their output representations (NeRF, 3DGS, pointmaps)?
- What are the key technical challenges driving innovation in feed-forward 3D scene modeling across different representation types?
- How do different design strategies for feature enhancement, geometry awareness, and efficiency translate across various 3D representations?

Core Idea

Traditional 3D reconstruction methods like NeRF and 3DGS require slow per-scene optimization, limiting practical deployment. Feed-forward methods solve this by learning to map images directly to 3D representations in a single forward pass, enabling fast generalization across scenes. The key insight is that despite diverse output formats (implicit fields, explicit primitives), most methods share similar architectural patterns and face common challenges. Rather than categorizing by representation type, this survey proposes a problem-driven taxonomy focusing on five core challenges: feature enhancement, geometry awareness, efficiency, data augmentation, and temporal modeling.

Key Formula

Pointmap coordinate transformation

$$X_{n,m} = P_m P_n^{-1} h(X_n)$$

X_n,m denotes pointmap X_n from camera n expressed in camera m's coordinate frame, P_m and P_n are world-to-camera poses, h is homogeneous mapping. This enables consistent 3D point representation across multiple views in feed-forward pointmap methods like DUSt3R.

Multi-objective feed-forward training loss

$$L = \sum_{t \in T} \lambda_t L_t(G, G^*, \hat{I}, I, P)$$

Combined loss function where T is set of loss terms, λ_t are weights, G is predicted 3D representation, G* is ground truth geometry, Î are rendered images, I are input images, P are poses. Enables joint training of encoder, decoder, and optional renderer components across diverse supervision signals.

Key Figure

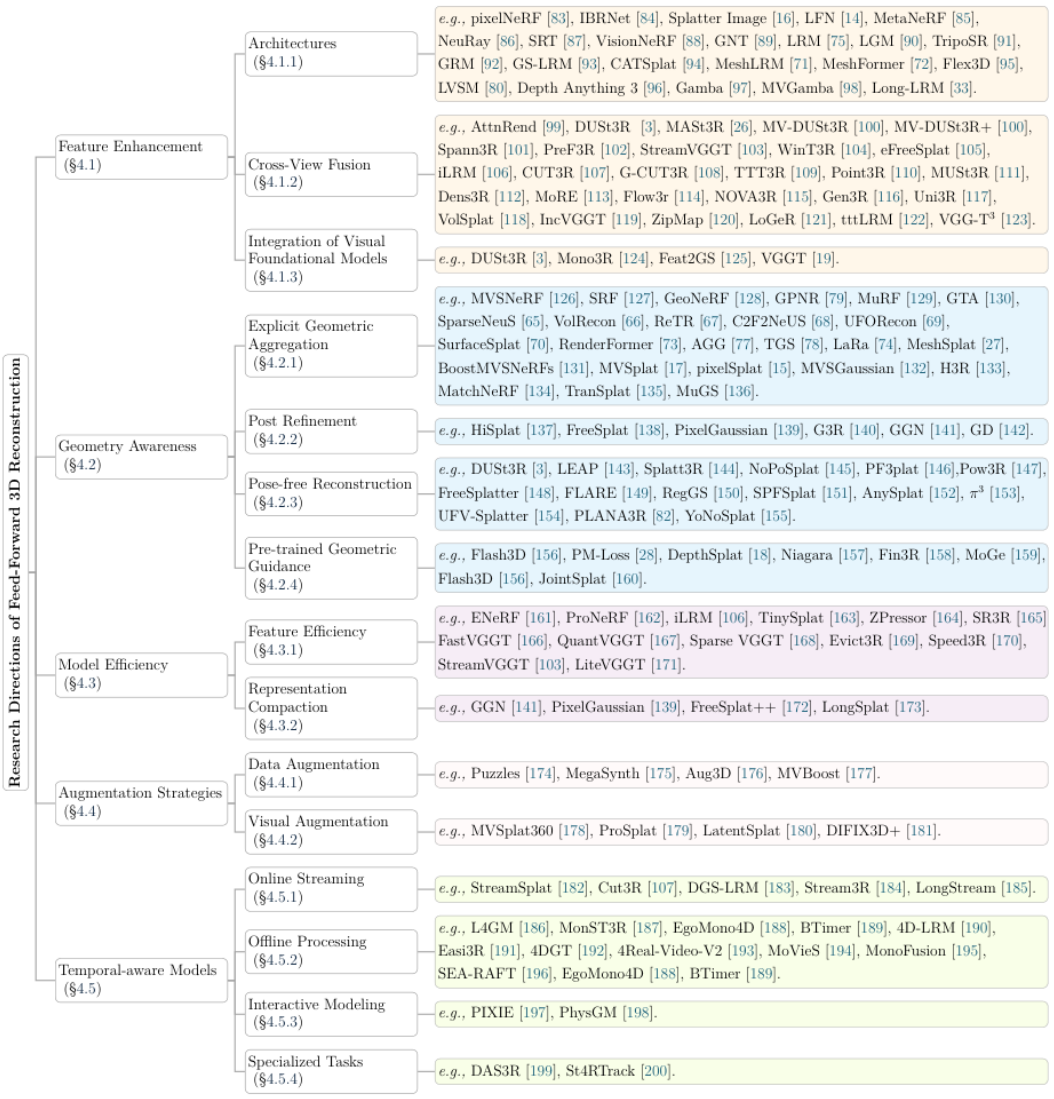

Figure 2. A taxonomy of feed-forward 3D reconstruction methods. This taxonomy summarizes the approaches discussed in the Directions section, organizing them into primary and secondary subcategories. For brevity, only representative methods are included. Here, we only consider NeRF, 3DGS and Point maps. Traditional feed-forward 3D reconstruction has also been explored with Voxel (3D ShapeNets [201], 3D-R2N2 [202]), mesh (Neural 3D Mesh Renderer [203], CMR [204], pixel2mesh [205], MeshGPT [206]), Occupancy (Occupancy Networks [5], Convolutional Occupancy Networks [64]), SDF (DeepSDF [4], MonoSDF [207]) and other representations.

Figure 2: Comprehensive taxonomy organizing 200+ feed-forward 3D methods into five problem-driven categories with specific subcategories and representative methods, providing the core organizational framework of the entire survey.

Limitations

- Survey nature limits novel technical contributions - provides organization and analysis rather than new methods or architectures
- Problem-driven taxonomy may still have overlapping categories where methods address multiple challenges simultaneously
- Limited quantitative comparison across methods due to inconsistent evaluation protocols and datasets in the field
- Rapid field evolution means some recent methods and trends may not be fully captured despite 395 references